

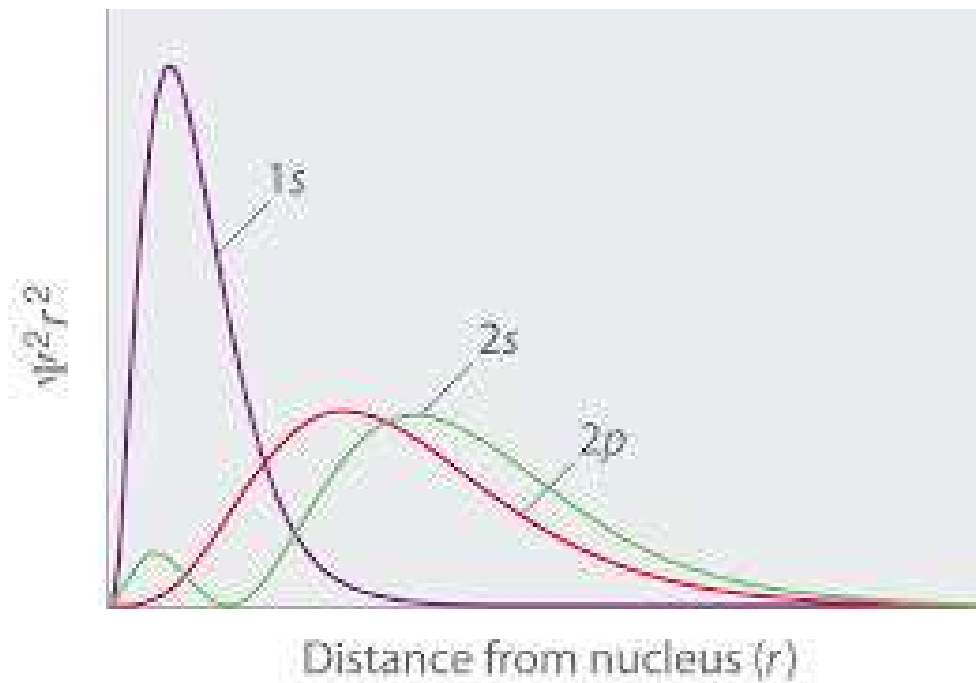
Practice Session 3

Assignment 1

1. How many radial nodes do 3p, 3d, 4f and 5s orbitals have? Show your work.

$$\text{No. of nodes} = n - l - 1$$

2. Draw the radial distribution functions for 2s and 2p orbitals of Hydrogen atom.



3. Calculate the Z^* for Na and Na^+ . Using this can you rationalize the trend in radii for the cation in comparison to neutral atom?

Na, atomic no. = 11

Electronic configuration $(1s^2) (2s^2 2p^6) (3s^1)$

$$\sigma_{\text{Na}} = (0 \times 0.35) + (8 \times 0.85) + (2 \times 1) = 8.8$$

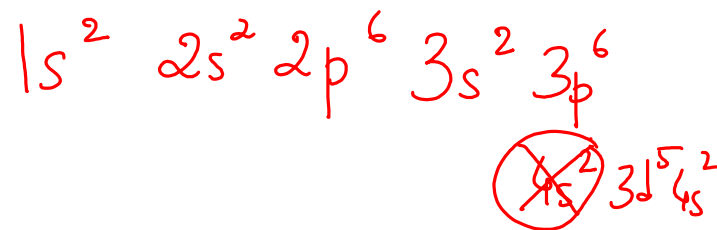
$$Z_{\text{Na}} = 11 - 8.8 = 2.2$$

$$\sigma_{\text{Na}^+} = (7 \times 0.35) + (2 \times 0.85) = 4.15$$

$$Z_{\text{Na}^+} = 11 - 4.15 = 6.85$$

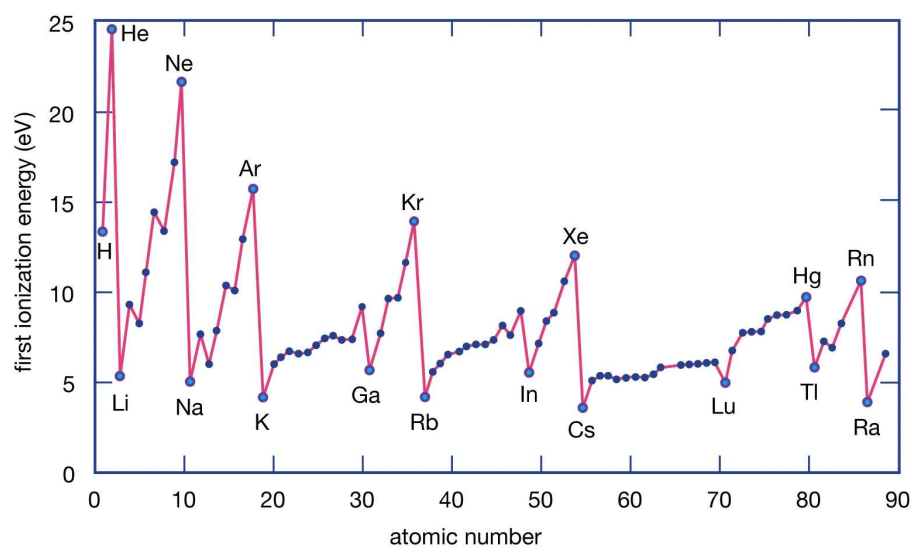
Since Z^* of $\text{Na}^+ > \text{Na}$, the size of $\text{Na}^+ < \text{Na}$

Mn

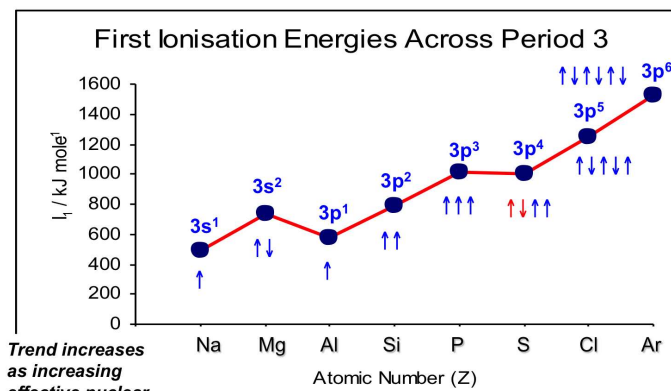


6

4. Account for the trends across Period 3 in (a) ionization energy,



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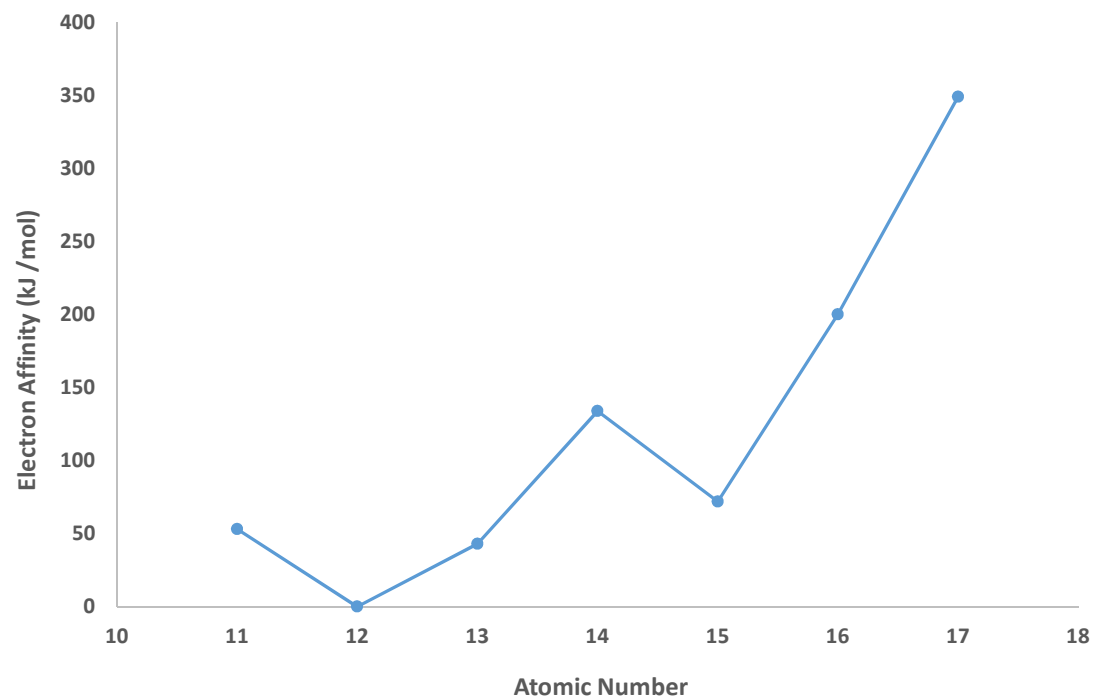


Trend increases
as increasing
effective nuclear
charge density

Decrease Al e⁻ in higher energy 3p orbital, less energy needed to lose e⁻

Decrease S as there is repulsion between the electron pair occupying the 3p orbital subshell in Sulfur

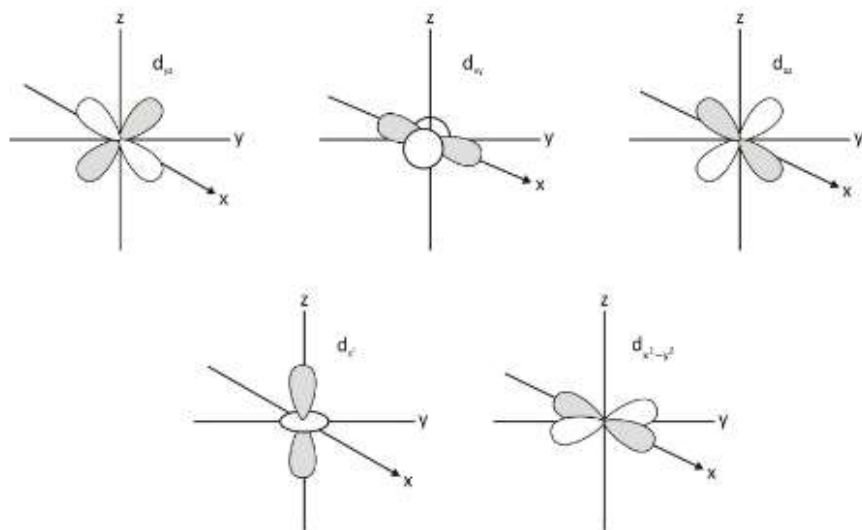
4. Account for the trends across Period 3 in (b) electron affinity.



5. Account for the fact that the two Group 5 elements niobium (Period 5) and tantalum (Period 6) have the same atomic radii.

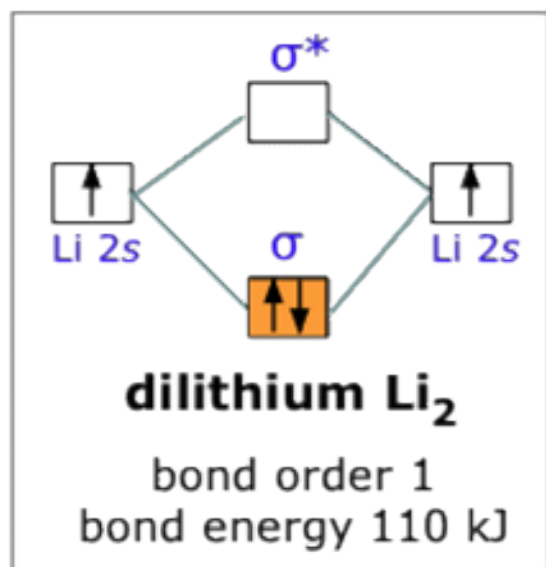
- Due to Lanthanide contraction.
- Electronic configuration of Nb is $[\text{Kr}] 4d^4 5s^1$
- Electronic configuration of Ta is $[\text{Xe}] 4f^{14} 5d^3 6s^2$
- Shielding effect of f electron is very poor. Due to presence of 4f electrons in Ta, the Z^* of Ta is much higher giving it the same size as the element in the previous period.

6. Draw the pictures of all the d orbitals. Label the Cartesian coordinate axes and the orbital lobes correctly.



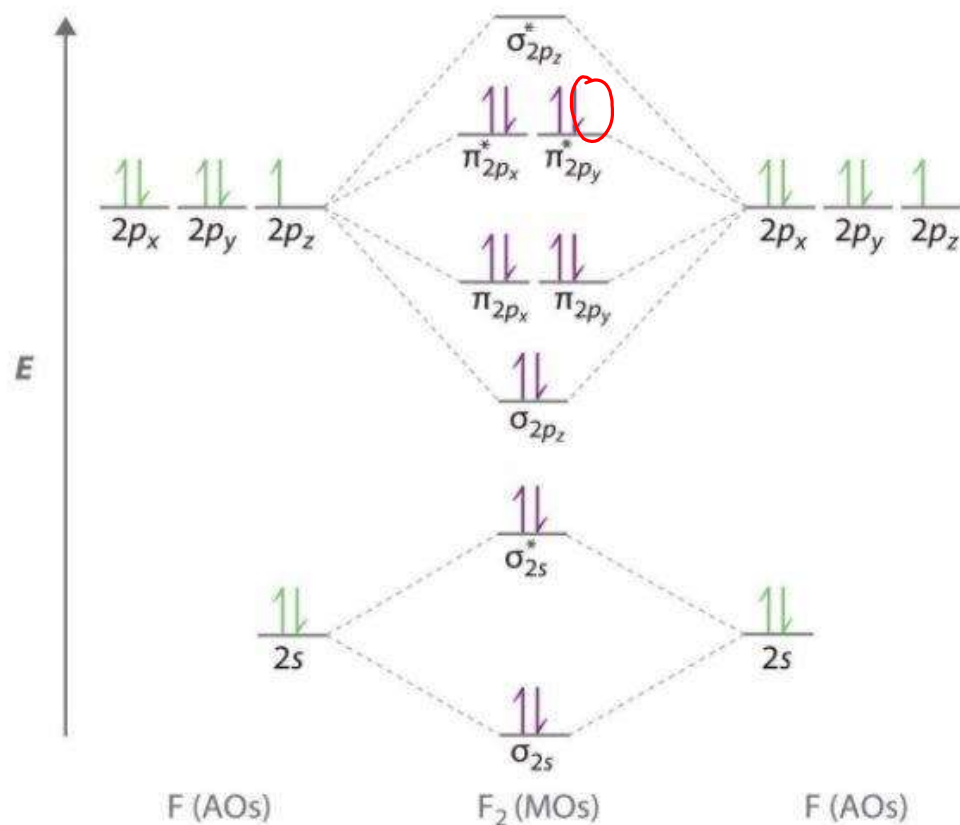
7. On the basis of molecular orbitals, predict the shortest bond and provide a brief explanation.

- Li_2^+ Li_2
- F_2^+ F_2



$\sigma_{1s}^2 \sigma_{1s}^{*2}$

Li_2



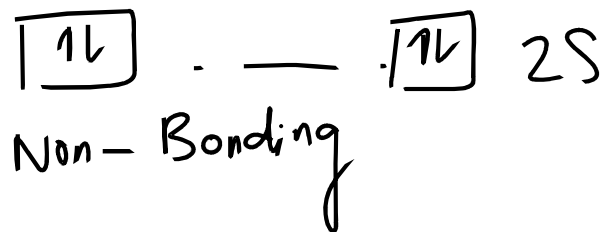
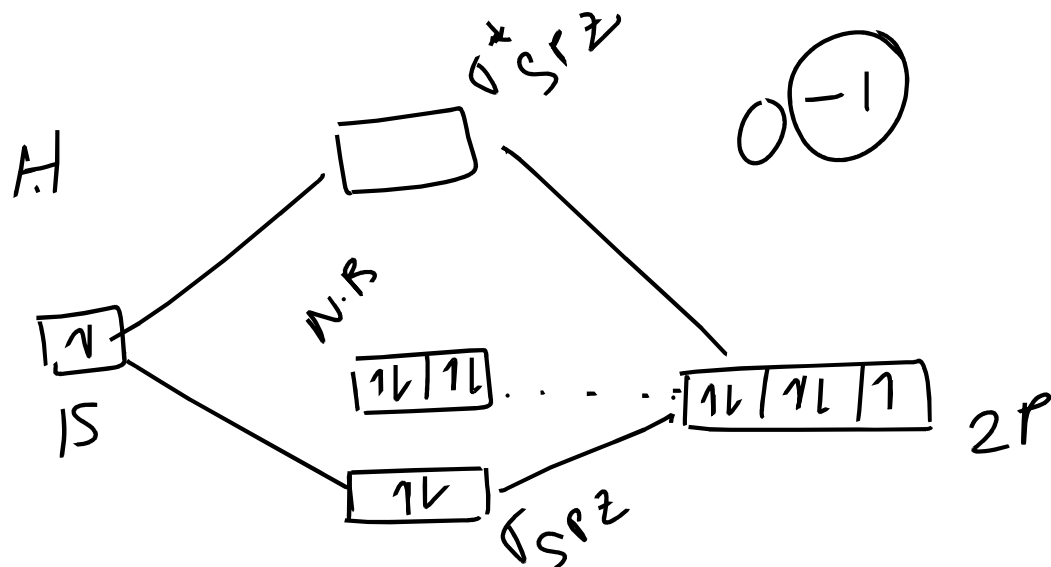
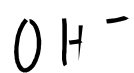
8. On the basis of molecular orbitals, predict the longest bond and provide a brief explanation.

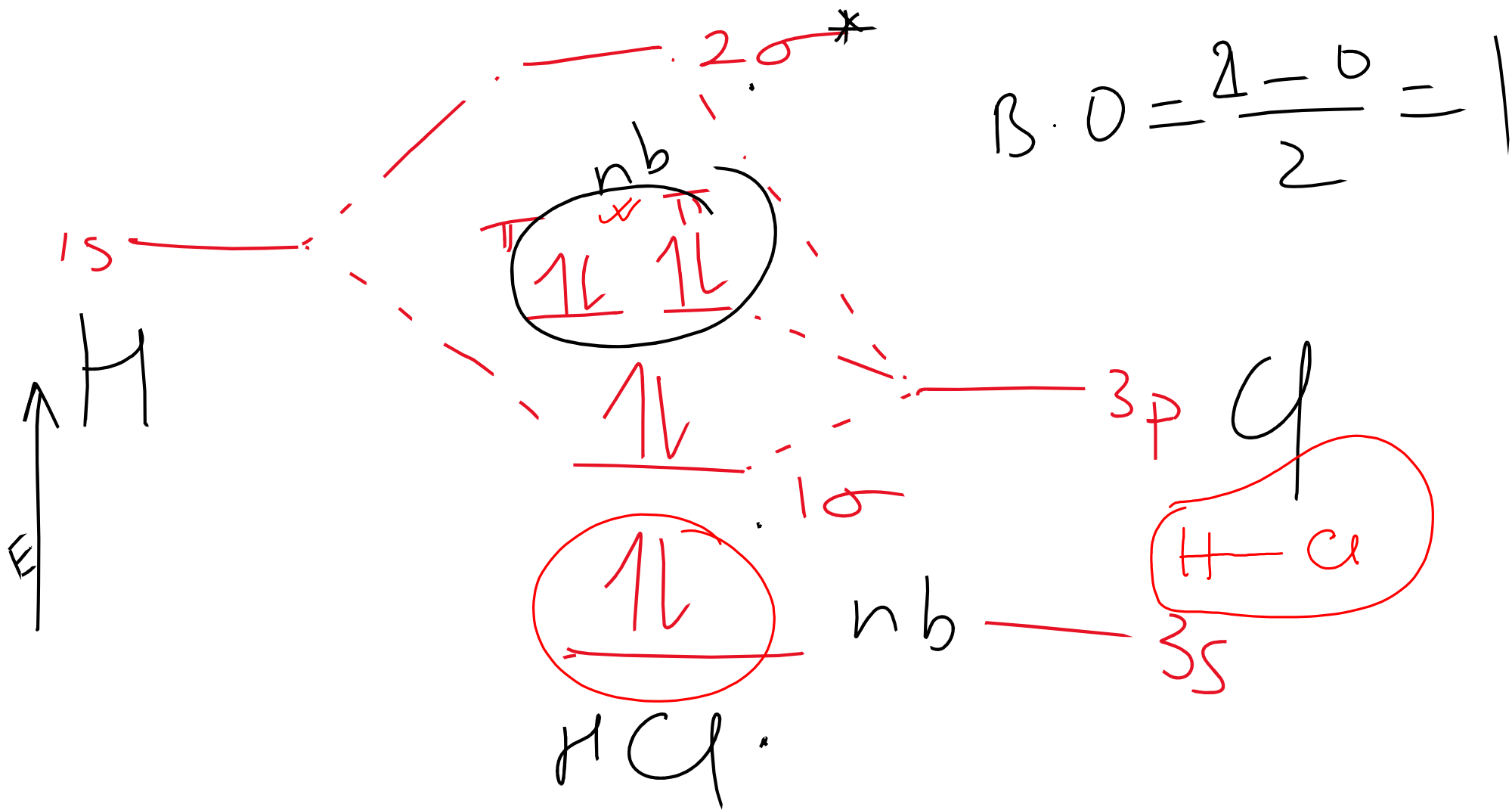


TABLE 5.2 Orbital Potential Energies

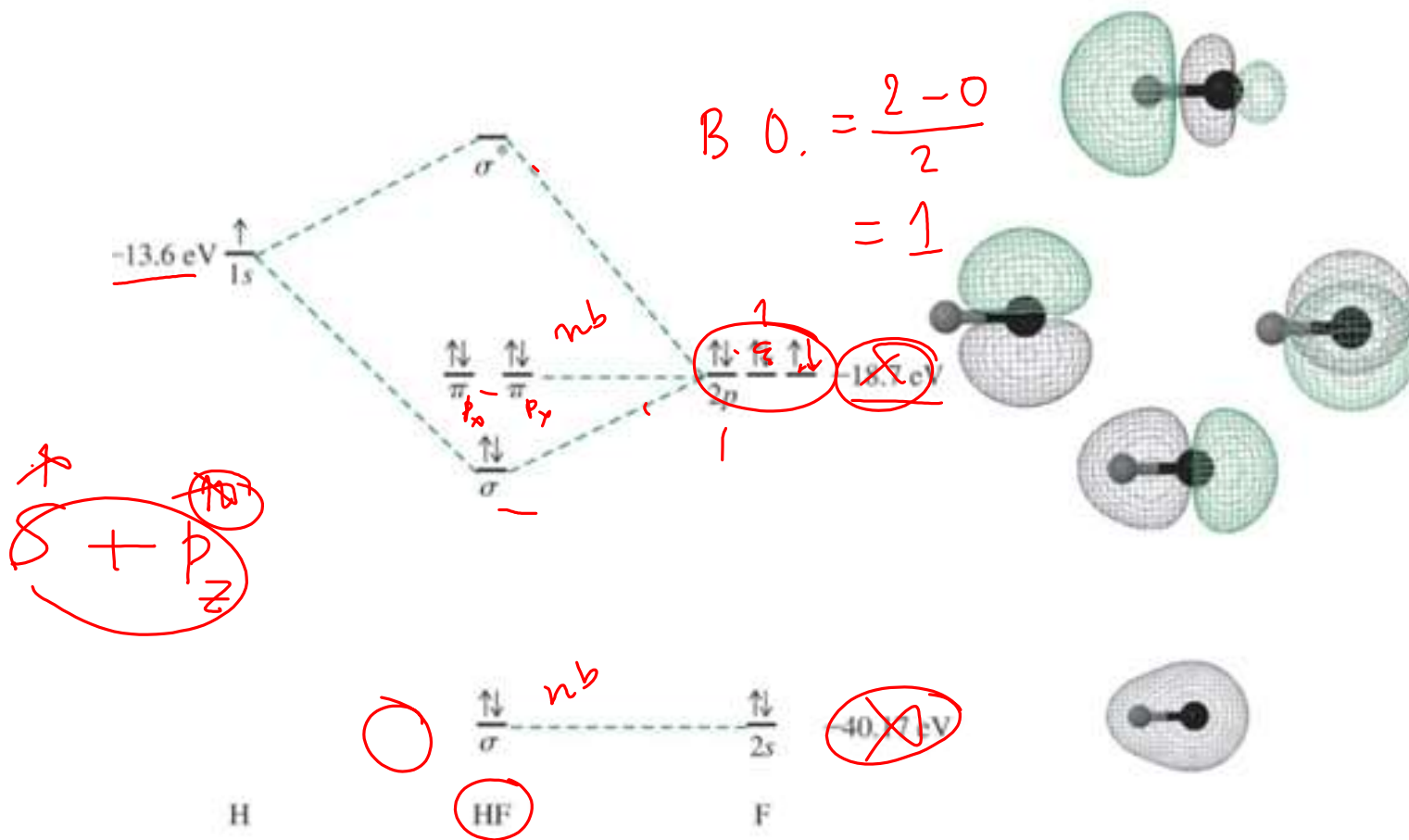
Atomic Number	Element	Orbital Potential Energy (eV)					
		1s	2s	2p	3s	3p	4s
1	H	-13.61					
2	He	-24.59					
3	Li		-5.39				
4	Be		-9.32				
5	B		-14.05	-8.30			
6	C		-19.43	-10.66			
7	N		-25.56	-13.18			
8	O		-32.38	-15.85			
9	F		-40.17	-18.65			
10	Ne		-48.47	-21.59			
11	Na				-5.14		
12	Mg				-7.65		
13	Al				-11.32	-5.98	
14	Si				-15.89	-7.78	
15	P				-18.84	-9.65	
16	S				-22.71	-11.62	
17	Cl				-25.23	-13.67	
18	Ar				-29.24	-15.82	

OH^-

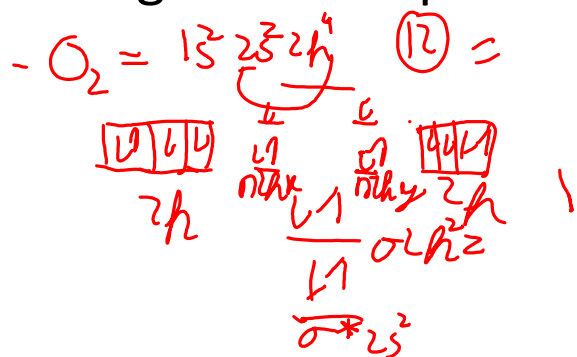




9. Draw the molecular orbital energy level diagram of HF molecule.



- Determine the bond order of the oxygen molecule, O_2 , the superoxide ion O_2^- , and the peroxide ion O_2^{2-} . Predict the bond length and strength of the species.

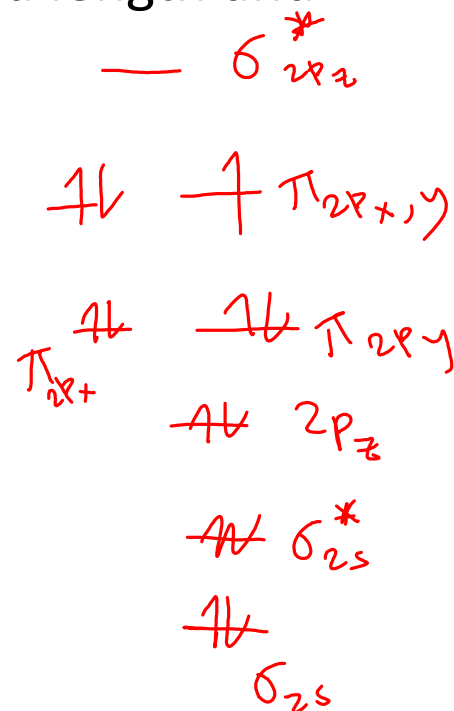


$$BO = \frac{6 - 2}{2} = 2$$

(O_2)

$$BO(O_2^-) = 1.5$$

$$BO(O_2^{2-}) = 1$$



- Draw the molecular orbital energy level diagram of HCl, OH⁻, CN⁻.
Calculate the bond order.